Blessed G/Michael Catholic Secondary School

Semester 1 Biology Final Examination for Grade11

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ Grade & Sec: Grade11 _____ R.No: _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** Pepsin preferentially hydrolyzes peptide bonds adjacent to aromatic amino acids.

____ **2.** The turnover number (kcat) of an enzyme is the number of substrate molecules converted per second when active sites are fully saturated.

____ **3.** The importance of adding hop flowers into the wort during beer brewing is to add sweet taste and sugar.

____ **4.** Deamination is the process occurring on excess amino acids in the liver to release the amino group.

____ **5.** Hormones are chemical signals that affect the physiological behavior of other target glands and tissues.

____ **6.** The modification of behavior resulting from individual life experiences is defined as learning.

II. Choose the best answer among the given alternatives.

____ **7.** Which biological property of DNA distinguishes it from RNA?

A. Double stranded
B. Double helix structure
C. Self-replicating capability
D. All of the above

____ **8.** When heterozygous yellow round seed pea plants (RrYy) are allowed to self-fertilize, the expected frequency of the RrYy genotype in the offspring is:

A. 1/16 **B.** 3/16 **C.** 4/16 **D.** 2/16

____ **9.** From the self-crossing of F1 dihybrids TtRr x TtRr, what is the probability of obtaining dwarf wrinkled (ttrr) offspring in F2? (T=tall, t=dwarf, R=round, r=wrinkled)

A. 1/8 **B.** 3/16 **C.** 1/16 **D.** 1/4

____ **10.** Which of the following is NOT a unique hallmark feature of meiosis?

A. Genetically identical daughter cells formation
B. Homologous synapsis formation (tetrads)
C. Reduction cell division (2n -> n)
D. Crossing over (genetic recombination)

____ **11.** Which component of a newly transcribed eukaryotic pre-mRNA is excised during RNA splicing before nuclear export and translation?

A. Introns
B. 5' 7-methylguanosine cap
C. Protein-coding exons
D. Ribosome binding site

____ **12.** An individual organism that carries two different alleles for a particular gene locus on homologous chromosomes is:

A. Recessive **B.** Homozygous **C.** Dominant **D.** Heterozygous

____ **13.** A man with blood type O marries a woman with heterozygous type A blood. What are the expected phenotypic probabilities for their children?

A. 25% chance of type A
B. 25% chance of type B
C. 50% chance of type A and 50% chance of type O
D. 75% chance of type A

____ **14.** Which of the following is NOT considered a direct primary application of modern genetics?

A. Commercial synthesis of recombinant human antibiotics
B. Agricultural crop enhancement and yield improvement
C. Forensic DNA fingerprinting for criminal investigation
D. Morphological sorting in traditional plant taxonomic classifications

____ **15.** When heterozygous dominant Bb and homozygous recessive bb are crossed, what percentage of progeny exhibits one of the parental genotypes?

A. 25% **B.** 75% **C.** 50% **D.** 100%

____ **16.** If the sequence of the coding (non-template) strand of DNA is 5'-AGCATGAC-3', what is the sequence of the transcribed mRNA?

A. 5'-UCGUACUG-3' **B.** 5'-UGGUACAG-3' **C.** 5'-AGCAUGAC-3' **D.** 5'-TCGTACTG-3'

____ **17.** According to Mendelian segregation, alleles occur paired in _____ and segregate during formation of _____.

A. Ovum, Zygote
B. Diploid somatic organisms, Gametes (ovules/pollen)
C. Dihybrid, Ovum
D. Polyploidy, Pollen

____ **18.** Which of the following enzymes plays NO direct role in the synthesis of proteins (transcription and translation)?

A. DNA Polymerase **B.** DNA Helicase **C.** RNA Polymerase **D.** Peptidyl transferase

____ **19.** In which type of enzyme inhibition does the inhibitor bind exclusively to the pre-formed enzyme-substrate (ES) complex?

A. Competitive inhibition
B. Non-competitive inhibition
C. Uncompetitive inhibition
D. Irreversible inhibition

____ **20.** Hibernation is an adaptive behavioral and physiological dormancy that allows animals to:

A. Adapt to extreme changes in seasonal temperature
B. Communicate territory boundaries with conspecifics
C. Defend geographic territories from predators
D. Conserve metabolic energy during prolonged winter food scarcity

____ **21.** Why can an ectotherm (poikilotherm) typically survive significantly longer without food than an endotherm of identical body mass?

A. Ectotherms possess substantially higher resting metabolic rates
B. Ectotherms do not expend metabolic calories generating internal body heat
C. Ectotherms dissipate more heat to the surroundings
D. Ectotherms possess specialized enzymatic storage tissues

____ **22.** Active synthesis of RNA and metabolic proteins required for cell growth primarily takes place during:

A. M phase **B.** S phase **C.** G1 phase **D.** G2 phase

____ **23.** During meiotic cell division, sister chromatids of individual chromosomes separate at:

A. Metaphase I **B.** Anaphase I **C.** Anaphase II **D.** Metaphase II

____ **24.** Mitotic spindle fibers and cytoskeletal microtubules are composed of polymeric subunits of:

A. Actin spindles **B.** Tubulin **C.** Flagellin **D.** Globulin

____ **25.** Which form of enzymatic inhibition is characterized by a decrease in the Michaelis constant (Km)?

A. Competitive inhibition
B. Non-competitive inhibition
C. Uncompetitive inhibition
D. Allosteric feedback activation

____ **26.** The specific class of digestive enzymes that hydrolyzes starch polymers into maltose sugars is:

A. Hydrolases **B.** Amylases **C.** Lipases **D.** Nucleases

____ **27.** Which pairing correctly reflects the biochemical composition of an Apoenzyme and a Coenzyme respectively?

A. Protein moiety and Organic vitamin derivative
B. Vitamin and Inorganic mineral ion
C. Amino acid and Mineral ion
D. Globular protein and Transition metal

____ **28.** In competitive enzyme inhibition, the inhibitor molecule bears a close structural resemblance to the:

A. Co-enzyme **B.** Cofactor **C.** Prosthetic group **D.** Substrate

____ **29.** The specific temperature at which an enzyme catalyzes a reaction at its maximum possible velocity is called the:

A. Catalytic temperature
B. Optimum temperature
C. Denaturation temperature
D. Basal physiological temperature

____ **30.** Which chemical bond type is NOT involved in transient enzyme-substrate catalytic binding?

A. Hydrogen bonds
B. Covalent disulfide bonds
C. Ionic attractions
D. Van der Waals dispersion forces

____ **31.** When human core body temperature drops below normal homeostatic levels, which physiological response does NOT occur?

A. Peripheral vasoconstriction
B. Increased cellular metabolic rate (shivering)
C. Piloerection (hairs raised)
D. Profuse sweating and diaphoresis

____ **32.** Homeostasis in biological organisms is best defined as:

A. Maintaining an unchangeable, rigid state of body chemistry
B. Static equilibrium with the external environment
C. Maintaining a dynamically stable, near-constant internal physiological state
D. Altering internal chemistry to match external fluctuations

____ **33.** Social organization within an animal group where specific ranked members enjoy preferential access to food and mates is called:

A. Agonistic display **B.** Territoriality **C.** Dominance hierarchy **D.** Altruism

____ **34.** Which distinct form of animal learning occurs exclusively during a rapid, critical sensitive window immediately after hatching or birth?

A. Associative conditioning
B. Imprinting
C. Habituation
D. Classical conditioning

____ **35.** Which primary germ layer forms outermost first during early embryonic gastrulation?

A. Ectoderm
B. Endoderm
C. Mesoderm
D. Both Endoderm and Mesoderm

____ **36.** Competitive enzymatic inhibition exhibits all of the following kinetic features EXCEPT:

A. The inhibitor's molecular conformation resembles that of the natural substrate
B. The inhibitor binds reversibly to the catalytic active site
C. Substrate and inhibitor compete directly for the active pocket
D. Maximum reaction velocity (Vmax) is permanently reduced

____ **37.** Reversible non-competitive enzyme inhibition includes all of the following characteristics EXCEPT:

A. The inhibitor binds to an allosteric site separate from the catalytic center
B. The inhibitor lacks structural homology to the substrate

C. The inhibitor binds to both free enzyme (E) and enzyme-substrate complex (ES)
D. The Michaelis constant (Km) is substantially increased

____ **38.** The vital physiological functions of the mammalian placenta include:

A. Diffusing respiratory oxygen from maternal blood to the fetal circulation
B. Removing metabolic urea and carbon dioxide from the fetal bloodstream
C. Providing physical shock absorption (role of amniotic fluid)
D. Both A and B (transport of nutrients/oxygen and waste elimination)